

ARIMA model (AutoRegressive Integrated Moving Average)

- The ARIMA model (AutoRegressive Integrated Moving Average) is one of the most widely used statistical methods for time series forecasting. It captures patterns such as trends, seasonality (if extended), and autocorrelation in data observed over time.
- **ARIMA(p, d, q) has three components:**
 - AR (AutoRegressive) – p
Uses past values of the series to predict the current value.
 - I (Integrated) – d
Represents the number of times the data is differenced to make it stationary.
 - MA (Moving Average) – q
Uses past forecast errors to improve predictions.
- **Time series data often shows:**
 - Trend (upward/downward movement)
 - Seasonality
 - Noise
- **ARIMA helps by:**
 - Making data stationary
 - Modeling relationships between past values and errors
 - Producing accurate forecasts

Understanding each component of ARIMA

1. AutoRegressive (AR) Model

The AR(p) model uses previous values:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \epsilon_t$$

◆ Meaning of each term

◆ 1. Y_t

- Value of the time series at **current time t**
 - This is what we are trying to predict
-

◆ 2. c (Constant / Intercept)

- A **baseline value**
- Think of it as the average level of the series
- If all other terms are zero, $Y_t = c$

👉 Example:

If $c = 10$, the model assumes data revolves around 10

◆ 3. $\phi_1, \phi_2, \dots, \phi_p$ (AR coefficients)

- These are **weights of past values**
- They tell how much influence past observations have



👉 Example:

- $\phi_1 = 0.8 \rightarrow$ strong influence of previous value
- $\phi_2 = 0.1 \rightarrow$ small influence of second lag

👉 Interpretation:

“How much should I trust past values to predict the present?”

◆ 4. $Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}$

- Past observations (lags)
 - These are actual previous data points
-

◆ 5. ϵ_t (Error term / noise)

- Random error at time t
- Represents **unpredictable part**

👉 Assumption:

- Mean = 0
- Constant variance



👉 Interpretation:

- Current value depends on past values

2. Integrated (I) - Differencing

To remove trend and make the series stationary:

$$Y'_t = Y_t - Y_{t-1}$$

Example:

Original data:

10, 12, 15, 18, 22 (trend present)

After differencing:

2, 3, 3, 4 (more stable)

👉 If needed, differencing is applied multiple times (d times).

3. Moving Average (MA)

The MA(q) model uses past errors:

$$Y_t = c + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$$

◆ Moving Average (MA) part

$$Y_t = c + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \dots + \theta_q\epsilon_{t-q}$$

◆ $\theta_1, \theta_2, \dots, \theta_q$

- Weights of **past errors**
- Help correct **previous mistakes**

👉 Interpretation:

“If I made an error before, adjust the prediction now.”

👉 Interpretation:

- Adjusts prediction using past mistakes

4. Combined ARIMA Model

Putting all together:

ARIMA(p,d,q)ARIMA(p, d, q)ARIMA(p,d,q)

Example:

- ARIMA(1,1,1)
 - 1 lag of past value
 - 1 differencing
 - 1 lag of error

Flow of ARIMA modeling

Step 1: Check stationarity

- Use plots or statistical tests
- If not stationary → apply differencing

Step 2: Identify p and q

Using:

- ACF (AutoCorrelation Function)
- PACF (Partial AutoCorrelation Function)

Conceptual ACF/PACF patterns:

AR model:

PACF: cuts off sharply

ACF: tails off gradually

MA model:

ACF: cuts off sharply

PACF: tails off gradually

Example:

Forecasting idea

Past Data → Model learns → Future Prediction

[10, 12, 13, 15, 18] → ARIMA → [20, 22, 23]

ARIMA is used when:

- Data is univariate
- Data has trend but no strong seasonality
- Observations are time-dependent

Limitations of ARIMA

- Not good for strong seasonality (use SARIMA instead)
- Requires careful parameter tuning
- Assumes linear relationships

◆ Extensions

- SARIMA → Seasonal ARIMA
- ARIMAX → Includes external variables
- SARIMAX → Seasonal + external variables

Simple real-life example

Predict:

- Stock prices
- Temperature
- Electricity demand

Example:

Past electricity demand → ARIMA → Future load forecast

NOTE:

Think of ARIMA as:

“A smart combination of past values + past changes + past errors to predict the future.”

◆ Question (10 Marks)

Consider a time series modeled using an **ARIMA(1,0,1)** model:

$$Y_t = 5 + 0.7Y_{t-1} + \epsilon_t + 0.4\epsilon_{t-1}$$

At time $t - 1$, the following values are known:

- $Y_{t-1} = 20$
 - $\epsilon_{t-1} = -2$
 - Current error $\epsilon_t = 1$
-

◆ Tasks:

1. Compute the value of Y_t .
2. Interpret the role of coefficients 0.7 and 0.4.
3. If $\epsilon_t = 0$, how does the prediction change?

◆ Solution

✓ Step 1: Substitute values

$$Y_t = 5 + 0.7(20) + 1 + 0.4(-2)$$

✓ Step 2: Calculate

- $0.7 \times 20 = 14$
- $0.4 \times (-2) = -0.8$

$$Y_t = 5 + 14 + 1 - 0.8 = 19.2$$

👉 Final Answer: $Y_t = 19.2$

✓ Step 3: Interpretation

- **0.7 (AR coefficient)**
 - Strong dependence on previous value
 - 70% of past value influences present
 - **0.4 (MA coefficient)**
 - Adjusts prediction using past error
 - Helps correct previous mistake
-

✓ Step 4: If $\epsilon_t = 0$

$$Y_t = 5 + 14 + 0 - 0.8 = 18.2$$

👉 Prediction decreases because **current randomness is removed**

ARIMA

Model Explained

ARIMA stands for **AutoRegressive Integrated Moving Average**.

ARIMA is used for **Time Series Forecasting**.

AR(P): AutoRegressive

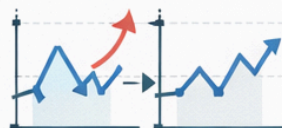
Lagged Values.



Uses past values to predict future values.

I(D): Integrated

Δ Differencing

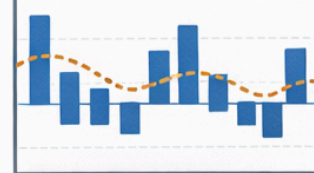


$$\Delta Y_t = Y_t - Y_{t-1}$$

Makes the series stationary.

MA(Q): Moving Average

Error Terms



Uses past forecast errors in the model.

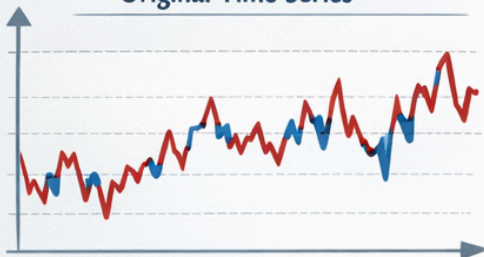
ARIMA Model: (p, d, q)

p = Number of AR terms.

d = Degree of Differencing.

q = Number of MA terms.

Original Time Series



Stationary Series After Differencing

